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Oefeningenreeks 4A nr. 62.12

$$\int \frac{x^2}{\sqrt{4-x^2}}dx \overset{x=2\sin t}{=} \int \frac{4\sin^2 t}{2\cos t} \cdot 2\cos t dt = \int 4\sin^2 t dt$$
$$= 4\int \frac{1-\cos 2t}{2}dt = 2\int dt - 2\int \cos 2t dt = 2t - \sin 2t + c$$
$$\overset{t=\operatorname{Bgsin}\left(\frac{x}{2}\right)}{=} 2\operatorname{Bgsin}\left(\frac{x}{2}\right) - \frac{x\sqrt{4-x^2}}{2} + c$$

References